Lab 1: OKE Quick Create with Managed Nodes

OKE cluster creation in the OCI Console, plus access from OCI Cloud Shell

Setting	Value
Region	ap-hyderabad-1 (Hyderabad)
Compartment	Root tenancy compartment
Workflow	Oracle Kubernetes Engine (OKE)
Node type	Managed nodes

Before you begin: Make sure you have permission to create OKE clusters, Compute instances, VCN resources, subnets, load balancers and related networking resources.

Part A: Create the cluster with Quick Create

Use this workflow when you want OCI to create the required networking resources automatically.

Step 1: Open the OKE cluster page

1. Sign in to the OCI Console.
2. Select the region **Hyderabad / ap-hyderabad-1**.
3. Open the navigation menu.
4. Go to **Developer Services > Containers & Artifacts > Kubernetes Clusters (OKE)**.
5. Select the required compartment.
6. Click **Create cluster**.

Step 2: Select Quick Create

1. Select **Quick create**.
2. Click **Proceed**.

Step 3: Configure the cluster

Enter or select:

- **Name:** for example, demo-oke-managed
- **Compartment:** select the required compartment.
- **Kubernetes version:** select the recommended available version.
- **Kubernetes API endpoint:** Public endpoint for a simple demonstration, or Private endpoint for a more secure production design.

Tip: For a first-time demonstration, a public endpoint is simpler (and is required for the Cloud Shell steps in Part C unless you attach Cloud Shell to the private network). For production, use a private endpoint with a bastion host or another approved access method.

Step 4: Configure networking

Quick Create automatically creates the network resources, including a VCN and subnets. Choose the worker node subnet visibility:

- **Public subnet:** easier for a basic demonstration.
- **Private subnet:** recommended for production.

OCI creates the required network components, such as:

- VCN
- API endpoint subnet
- Worker node subnet
- Load balancer subnet
- Route tables
- Internet gateway
- NAT gateway
- Security lists

The load balancer subnet is generally created as a public subnet in the Quick Create workflow.

Step 5: Select managed nodes

In the node configuration section:

1. For **Node type**, select **Managed**.
2. Select a **Shape**, such as a small VM shape available in your tenancy.
3. Select an OKE-compatible **Image**.
4. Set the **Number of nodes**: 1 for a simple demo, or 2 or more to test availability.
5. Select whether the nodes use a public or private subnet.

> **Note:** For a trial tenancy, check the available shapes and service limits before selecting the node count.

Step 6: Configure optional managed-node settings

Expand the advanced options if required:

- Boot volume size
- Boot volume encryption
- SSH public key
- Kubernetes node labels
- Additional node settings

> **Important:** Add your SSH public key if you need SSH access to the worker nodes. Never enter a private SSH key.

Step 7: Review and create

1. Review the cluster, networking and node pool settings.
2. Confirm that **Managed** is selected as the node type.
3. Click **Create cluster**.
4. Wait for the cluster and node pool to reach **Active** / **Running** status.

> **Note:** Quick Create is suitable for a demonstration because it creates the supporting network automatically. It provides fewer choices for custom subnets, capacity types and advanced networking.

Part B: Verify the cluster in the Console

1. Open the cluster from the Kubernetes Clusters list.
2. Go to **Node pools**.
3. Confirm the managed node pool is **Active**.
4. Open the node pool and verify that the nodes are running.

Recommended demo configuration

- Public API endpoint
- Public worker node subnet
- Managed nodes
- One small VM node
- Default networking
- No custom encryption
- Optional SSH public key

This is the easiest configuration for a classroom or demonstration environment.

Part C: Access the cluster from OCI Cloud Shell

OCI Cloud Shell is a browser-based terminal that is pre-authenticated with your OCI identity and comes with the OCI CLI and **kubectl** already installed, so no local setup is needed.

> **Prerequisites:** The cluster is **Active**; you have IAM permission to use the cluster (for example, manage cluster-family in the compartment); and, for a private API endpoint, Cloud Shell must be on a private network that can reach the endpoint subnet (see Step C6).

Step C1: Open the cluster's Access Cluster dialog

1. In the OCI Console, open **Developer Services > Containers & Artifacts > Kubernetes Clusters (OKE)**.
2. Click the name of your cluster (for example, demo-oke-managed).
3. On the cluster details page, click **Access cluster**.
4. Select **Cloud Shell Access**.

Step C2: Launch Cloud Shell

1. Click **Launch Cloud Shell** (or the Cloud Shell icon in the top-right of the Console header).
2. Wait for the terminal to start. Confirm the region banner shows the correct region.

Step C3: Generate the kubeconfig file

Copy the command shown in the **Access Your Cluster** dialog, **Step 2**, and paste it into Cloud Shell. It looks like this (your cluster OCID will differ):

```
oci ce cluster create-kubeconfig \
  --cluster-id ocid1.cluster.oc1.ap-hyderabad-1.<unique_ID> \
  --file $HOME/.kube/config \
  --region ap-hyderabad-1 \
  --token-version 2.0.0 \
  --kube-endpoint PUBLIC_ENDPOINT
```

> **Note:** Use **--kube-endpoint PRIVATE_ENDPOINT** if your cluster uses a private API endpoint. Always copy the exact command from the Console so the cluster OCID is correct.

Expected output: *New config written to the Kubernetes configuration file: /home/<user>/.kube/config*

Step C4: Verify kubectl access

```
kubectl get nodes
```

The output should show the managed worker nodes in the **Ready** state. Then check the system pods:

```
kubectl get pods -A
kubectl cluster-info
```

Step C5: Optional test deployment

```
kubectl create deployment nginx-demo --image=nginx
kubectl get pods
kubectl expose deployment nginx-demo --type=LoadBalancer --port=80
kubectl get svc nginx-demo --watch
```

When EXTERNAL-IP changes from <pending> to an IP address, browse to **http://<EXTERNAL-IP>**. Clean up afterwards:

```
kubectl delete service nginx-demo
kubectl delete deployment nginx-demo
```

Step C6: Private API endpoint (only if applicable)

1. In Cloud Shell, click the **Network** menu (top of the Cloud Shell window) and choose **Ephemeral private network setup** (or select an existing private endpoint).
2. Select the cluster VCN and a subnet that can reach the API endpoint subnet; ensure security rules permit access to port 6443.
3. Re-run the create-kubeconfig command with **--kube-endpoint PRIVATE_ENDPOINT**, then re-run kubectl get nodes.

Troubleshooting

Symptom	Likely cause / fix
Unable to connect / timeout	API endpoint is private and Cloud Shell is on the public network, or security rules block port 6443. Use Step C6.
Forbidden / unauthorized	Missing IAM policy for the user or group to use the cluster. Ask an administrator.
Wrong cluster targeted	Re-run create-kubeconfig with the correct cluster OCID and check: kubectl config current-context
Token or session expired	Re-run the create-kubeconfig command; Cloud Shell sessions time out when idle.

Important checks before clicking Create

- The selected Compute shape is available in ap-hyderabad-1.
- Your trial tenancy has sufficient Compute and networking quota.
- You have saved the SSH public key if node access is required.
- Required IAM permissions are available.

Sources

https://docs.oracle.com/en-us/iaas/Content/ContEng/Tasks/contengcreatingclusterusingoke.htm
https://docs.oracle.com/en-us/iaas/Content/ContEng/Tasks/contengdownloadkubeconfigfile.htm